Jasdeep Sidhu

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Education

Doctorate Degree in Computer Science *Sep 2026 – Present*
University of California, Los Angeles | GPA: 4.0

Master's Degree in Computer Science *Sep 2024 – Jun 2026*
Stanford University | GPA: 3.98

Bachelor's Degree in Computer Science *Sep 2021 – Jun 2026*
Stanford University | GPA: 3.98

Related Coursework: Machine Learning Algorithms (COMSCI 260), Advanced Data Structures (CS 166), Networking (CS 144), Building AI Language Models from Scratch (CS 336), Quantum Computing(CS 259Q), Operating System Principles (CS 111), Combinatorial Optimization (CS 261), Advanced Cryptography (CS 355), Cryptography (CS 255), Blockchain and Cryptocurrencies (CS 251), Design and Analysis of Algorithms (CS 161), Modern Algorithmic Toolbox (CS 168), Programming Abstractions (CS 106B), Computer Organization and Systems (CS 107), Algebraic Error Correcting Codes (CS 250), Probabilistic Methods (Math 159), Theory of Computation (CS 154), Graduate Abstract Algebra (Math 210A), Algebraic Number Theory (MATH 154), Discrete Mathematics (MATH 60DM Series),

Honors Math Thesis *Sep 2024 – Jun 2025*
Stanford University

Worked with Professor Jan Vondrak on stable committee selection using different voter preference structures.

Preliminary Arizona Winter School *Fall 2023*
Completed a six-week intensive course on elliptic curves with complex multiplication at the Preliminary Arizona Winter School (2023), solving weekly problem sets and engaging in collaborative problem sessions on class field theory, L-functions, and Galois representations.

Directed Reading Program *Winter 2022*
Stanford University

Completed 10 weeks of directed reading on Janson, Luczak, and Rucinski's *Random Graphs* with a graduate student mentor.

Work Experience

Machine Learning Engineer Intern *Jun 2026 – Sep 2026*
Guava AI

- Built and shipped a Do Not Call list feature into the production voice-agent backend, designing the MongoDB schema and update logic to persist and enforce compliance state across live calls
- Designed and built a Do Not Call dashboard giving the compliance team visibility into companies flagged by callees as unwanted, backed by the MongoDB persistence layer
- Benchmarked candidate models for latency and accuracy trade-offs across the voice pipeline and implemented back-end/AI infrastructure improvements that reduced end-to-end response latency

Software Engineering Intern *Jan 2026 – Jun 2026*
AllHeart Health

- Built a patient-facing health dashboard (React frontend, Node.js API layer) displaying heart rate, sleep, blood pressure, appointments, and medication tracking, sourced from a Python/FastAPI voice-agent backend integrated with FHIR R4 EHR data

Cryptography Teaching Assistant *Jan 2026 – Jun 2026*
Stanford University

- TA for Stanford's CS 355 Advanced Cryptography class, covering elliptic-curve cryptography, sigma protocols, multi-party computation, and lattice cryptography
- TA for Stanford's CS 255 Cryptography class, covering symmetric-key encryption, public key cryptography, digital signatures, and zero-knowledge protocols
- Held office hours, graded homework, and managed course programming projects

Linear Algebra/Multivariable Calculus Teaching Assistant *Sep 2025 – Dec 2025*
Stanford University

- TA for Stanford's Math 51, the introductory linear algebra and multivariable calculus class for around 500 students
- Taught four hours of section per week, held office hours, and graded exams

Quantitative Trading Intern *Jun 2024 - Aug 2024*
IMC Financial Markets

- Implemented Black-Scholes option pricing models in Python and validated them against Monte Carlo simulation, applying the results to valuation-based market making activities
- Analyzed the price and volatility impact of early-session trades using Python and pandas, building an intern project around opening-trade market impact

CS106 Section Leader, Teaching Assistant *Sep 2022 – Mar 2023*
Stanford University

- TA for Stanford's introductory computer science class for over 1000 students
- Taught weekly discussion sections for about 20 students, graded assignments and exams, and debugged student C++ code involving recursion, pointers, and hash table implementations during weekly office hours

Logic Camp Teaching Assistant *Jun 2022 - Aug 2022*
Stanford University

- TA for logic classes at Stanford meant for high school and college students
- Worked 1-on-1 with students, facilitated discussions, and helped them understand propositional, relational, and functional logic and proof-writing

In Focus Program Participant *May 2022*
Jane Street

- Selected to participate in a program introducing quantitative trading, technology, and collaborative problem-solving in financial markets at Jane Street.

Projects

Discrete Math Researcher *May 2025 – Sep 2025*
DIMACS, Rutgers University

- Proved tight distortion bounds for randomized voting rules under bounded utility profiles, establishing matching $\Omega(\sqrt{d})$ lower and $O(\sqrt{d})$ upper bounds for approximately stable committee selection
- Developed dice-based utility constructions and analyzed Nash social welfare to characterize distortion-optimal randomized mechanisms, using matrix factorization to construct symmetric ranking distributions
- Selected to attend discrete math lectures at Charles University and the Midsummer Combinatorial Workshop Conference in Prague

Undergraduate Computer Science Researcher *Aug 2023 – Dec 2023*
Stanford University

- Worked under the mentorship of Professor Michael Bernstein to rigorously assess the autoformalization capabilities of LLMs
- Designed an evaluation benchmark for Lean4 and applied it to test state-of-the-art LLMs, including GPT-3.5, GPT-4, and Gemini Pro
- Co-authored a paper that appeared in the ICLR 2024 Tiny Papers Conference (`https://arxiv.org/pdf/2406.06555`)

Undergraduate Math Researcher *Jun 2023 - Sep 2023*
Georgia Institute of Technology

- Worked under the mentorship of Professor Tom Kelly to study statistical problems of Latin squares
- Co-authored a paper accepted into the *Journal of Combinatorial Designs* (`https://arxiv.org/pdf/2311.04152`)
- Presented the work at the Joint Mathematics Meeting (JMM) Conference

Math Talk *Spring 2022*
Stanford University

- Gave a 15-minute math talk on triangles in random graphs to other directed reading participants

Awards

NSF Graduate Fellow
National Science Foundation

Recipient of the NSF-GRFP, a prestigious national fellowship recognizing excellence in research and academic achievement.

Publications

Subsquares in Random Latin Squares and Rectangles *2026*
Alexander Divoux, Tom Kelly, Camille Kennedy, Jasdeep Sidhu
Journal of Combinatorial Designs, 34(4), 184–197.

An Evaluation Benchmark for Autoformalization in Lean4 *2024*
Aryan Gulati, Devanshu Ladsaria, Shubhra Mishra, Jasdeep Sidhu, Brando Miranda
ICLR 2024 Tiny Papers.

Skills

Languages: Python, C++, C, C#, Java, JavaScript, LaTeX, Rust, SQL
Frameworks/Tools: React, Node.js, FastAPI, MongoDB